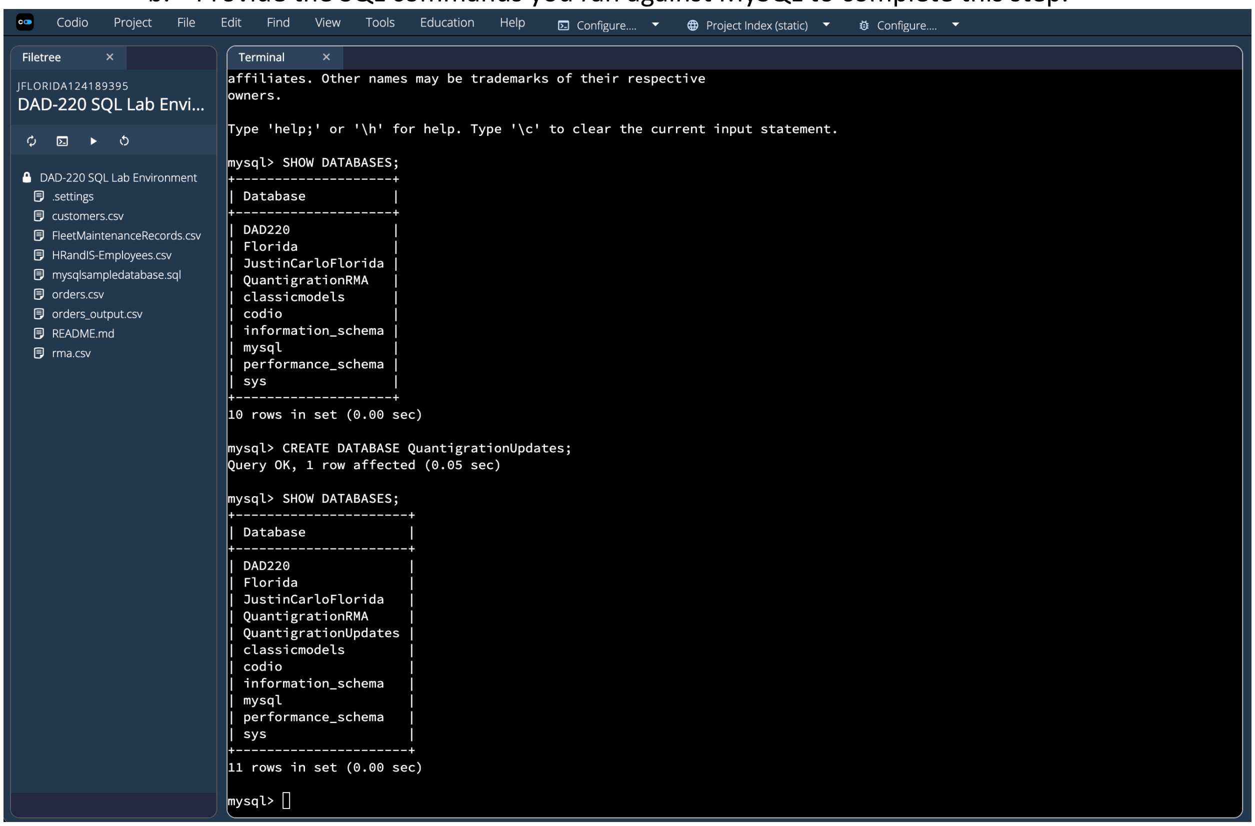
DAD 220 Project One Template

By: Justin Carlo Florida

Complete these steps as you work through the directions for Project One. Replace the bracketed text with your screenshots and brief explanations of the work they capture. Each screenshot and its explanation should be sized to approximately one-quarter of the page with the description written below the screenshot. Follow these rules for each of the prompts and questions below. Review the example document in the Supporting Materials section of the Project One Guidelines and Rubric for assistance.

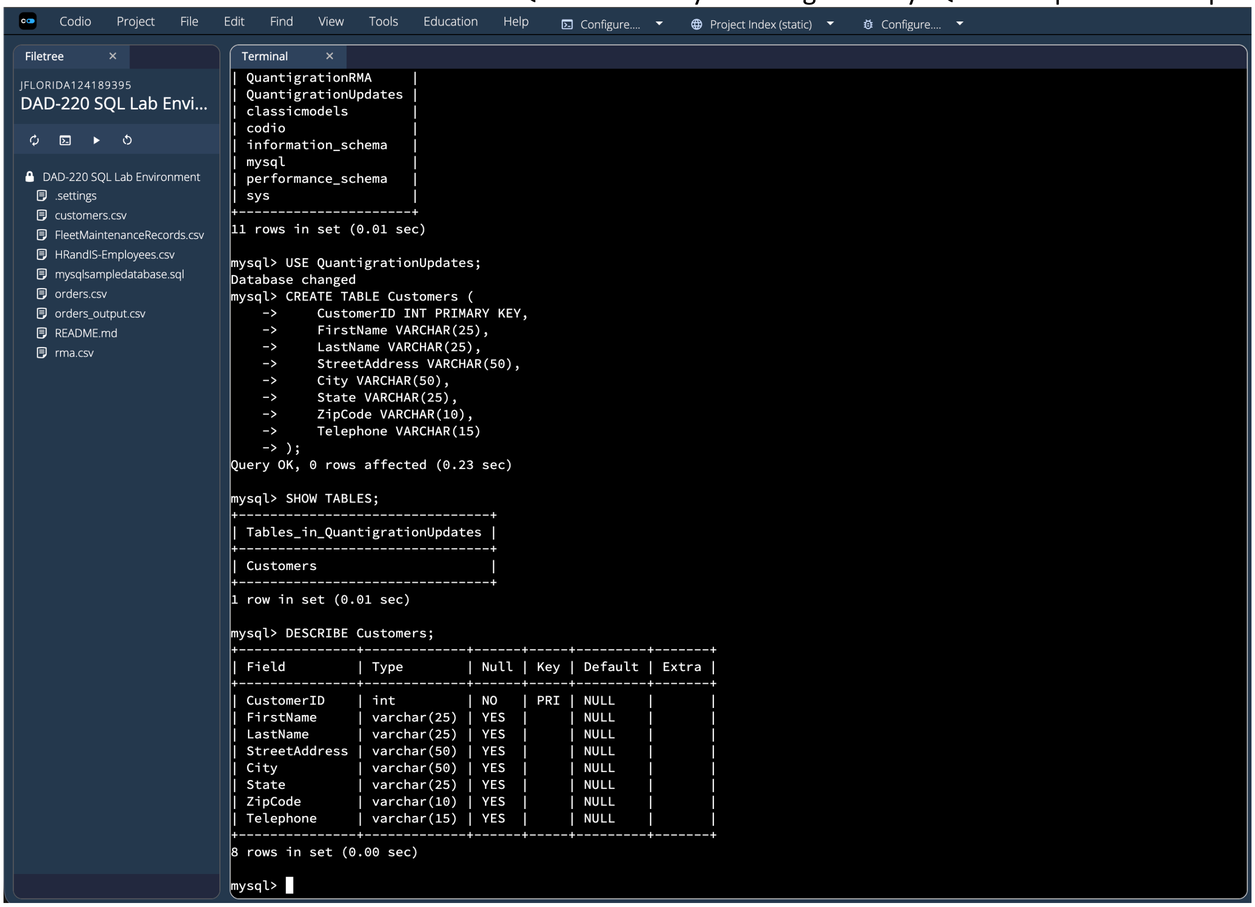
Step One: Create a Database

1. In your online IDE (Codio), **create a database schema** called QuantigrationUpdates that will hold tables by using SQL commands.
    a. List out the database name on the screen.
    b. Provide the SQL commands you ran against MySQL to complete this step.



I created a new database schema named QuantigrationUpdates using the CREATE DATABASE command in MySQL. The message “Query OK” confirms that the database was successfully created.

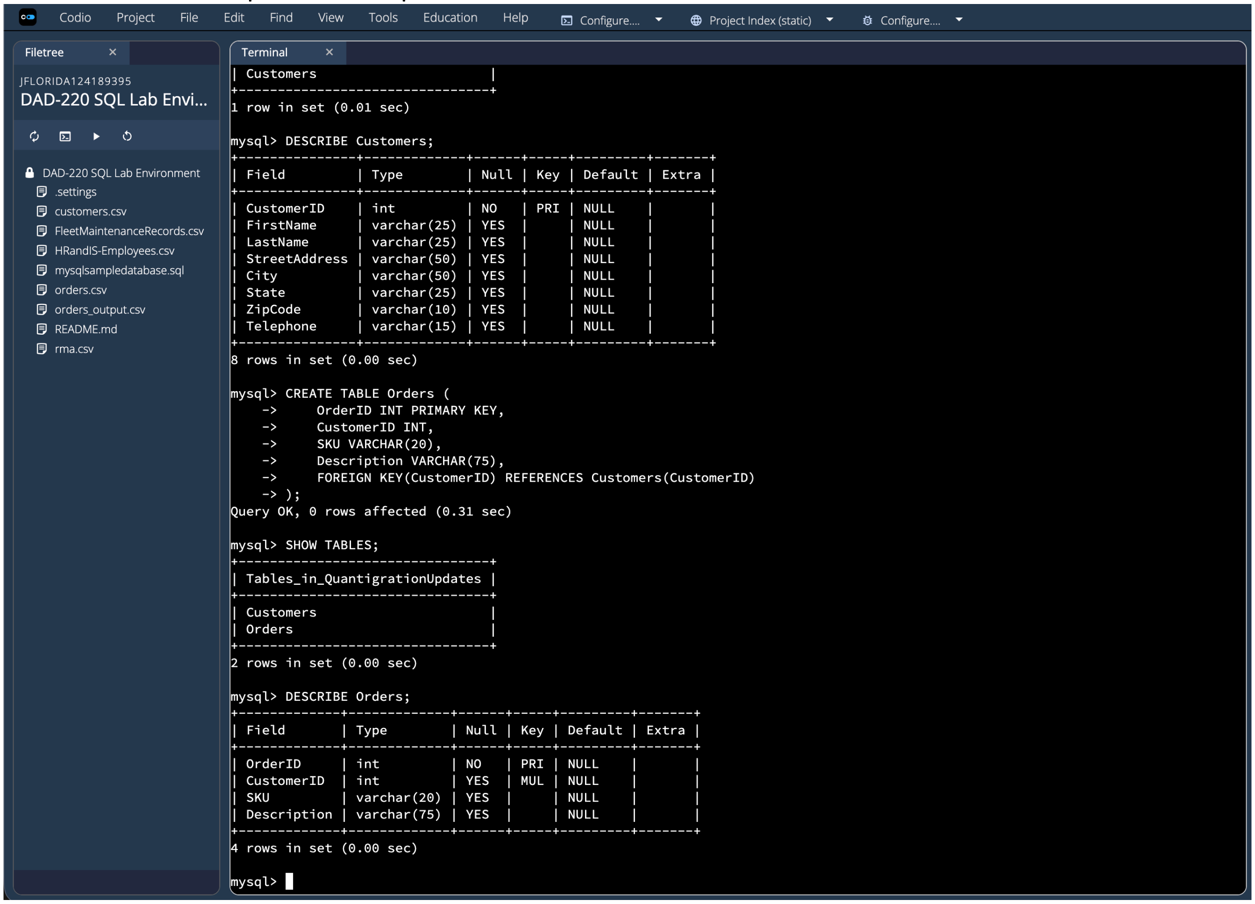
2. Connect to the QuantigrationUpdates schema. Using the ERD as a reference, **write SQL commands to create** the following **tables** with the appropriate attributes and keys to demonstrate relationships based on the ERD.
    a. A table named Customers to store customer information with a primary key of Customer ID. Provide the SQL commands you ran against MySQL to complete this step.




```
| QuantigrationRMA     |
| QuantigrationUpdates |
| classicmodels        |
| codio                |
| information_schema   |
| mysql                |
| performance_schema   |
| sys                  |
+----------------------+
11 rows in set (0.01 sec)

mysql> USE QuantigrationUpdates;
Database changed
mysql> CREATE TABLE Customers (
    ->     CustomerID INT PRIMARY KEY,
    ->     FirstName VARCHAR(25),
    ->     LastName VARCHAR(25),
    ->     StreetAddress VARCHAR(50),
    ->     City VARCHAR(50),
    ->     State VARCHAR(25),
    ->     ZipCode VARCHAR(10),
    ->     Telephone VARCHAR(15)
    -> );
Query OK, 0 rows affected (0.23 sec)

mysql> SHOW TABLES;
+--------------------------------+
| Tables_in_QuantigrationUpdates |
+--------------------------------+
| Customers                      |
+--------------------------------+
1 row in set (0.01 sec)

mysql> DESCRIBE Customers;
+---------------+-------------+------+-----+---------+-------+
| Field         | Type        | Null | Key | Default | Extra |
+---------------+-------------+------+-----+---------+-------+
| CustomerID    | int         | NO   | PRI | NULL    |       |
| FirstName     | varchar(25) | YES  |     | NULL    |       |
| LastName      | varchar(25) | YES  |     | NULL    |       |
| StreetAddress | varchar(50) | YES  |     | NULL    |       |
| City          | varchar(50) | YES  |     | NULL    |       |
| State         | varchar(25) | YES  |     | NULL    |       |
| ZipCode       | varchar(10) | YES  |     | NULL    |       |
| Telephone     | varchar(15) | YES  |     | NULL    |       |
+---------------+-------------+------+-----+---------+-------+
8 rows in set (0.00 sec)

mysql> 
```

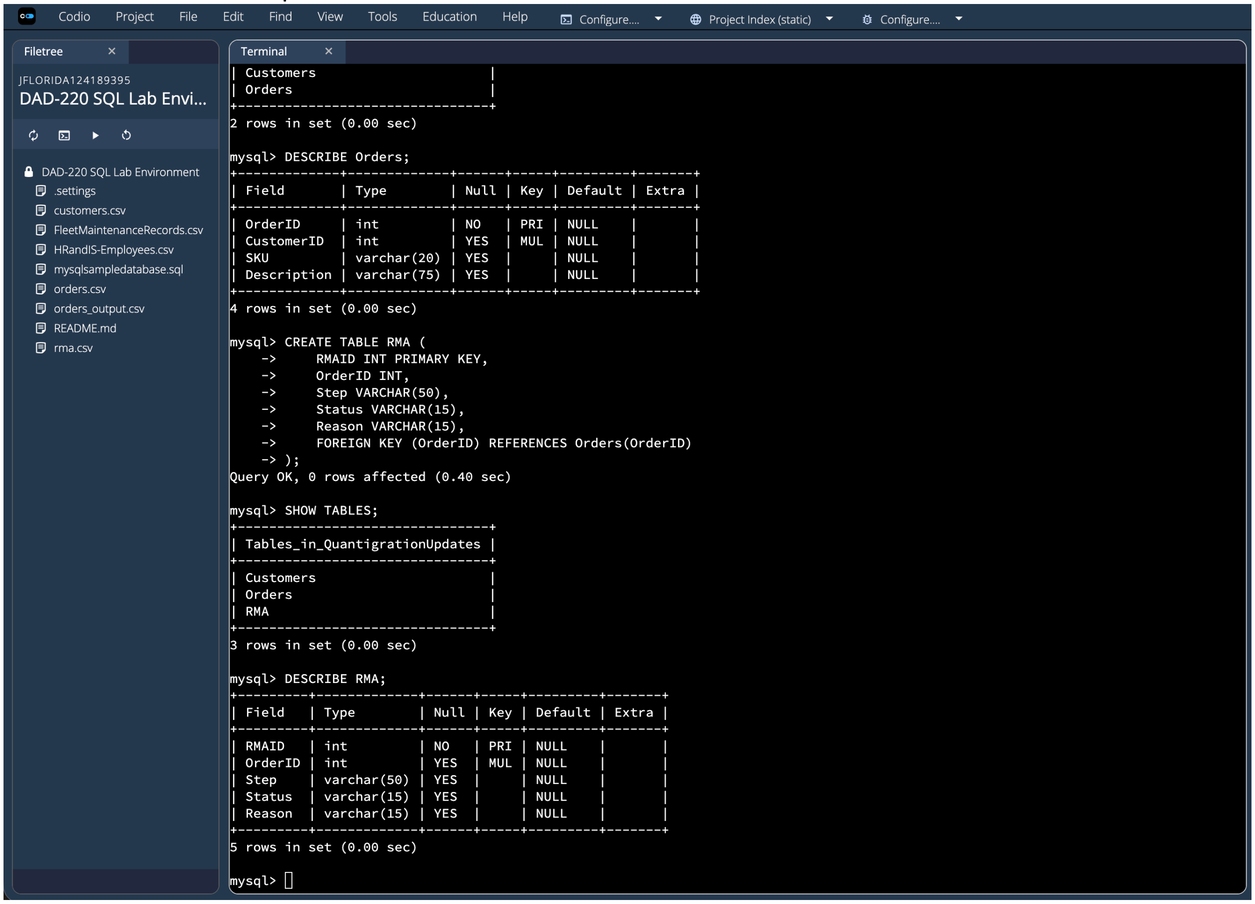
I connected to the QuantigrationUpdates database and created the Customers table using the ERD as a reference. The table includes CustomerID as the primary key and additional attributes such as first name, last name, street address, city, state, zip code, and telephone. I verified the table structure using the DESCRIBE Customers; command, which confirms that all columns were created correctly.

b. A table named Orders to store order information with a primary key of Order ID and a foreign key of Customer ID. Provide the SQL commands you ran against MySQL to complete this step.



I created the Orders table in the QuantigrationUpdates database based on the ERD. The table includes OrderID as the primary key and CustomerID as a foreign key to establish a relationship with the Customers table. This ensures that each order is linked to a valid customer. Additional attributes such as SKU and description were included to store order details. I verified the table creation using SQL commands to confirm the structure.

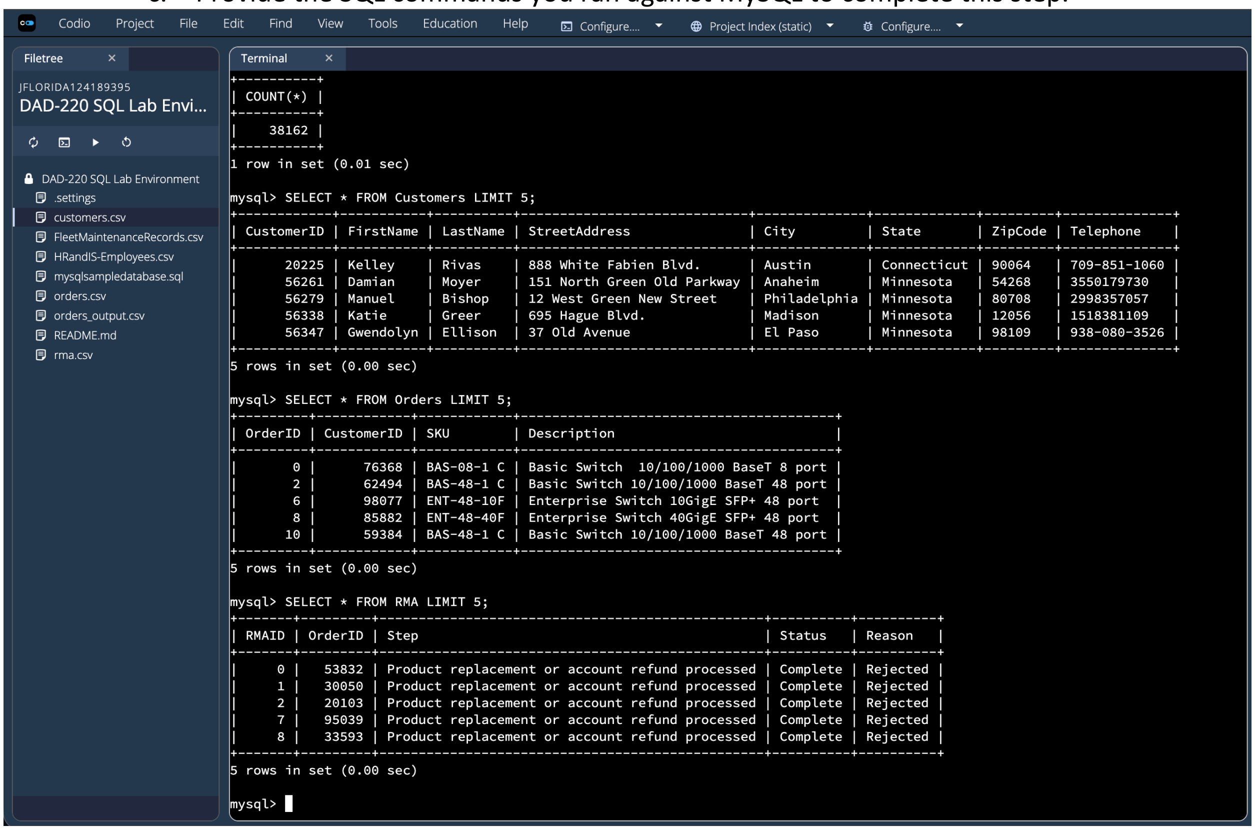
c. A table named RMA to store RMA information with a primary key of RMA ID and a foreign key of Order ID. Provide the SQL commands you ran against MySQL to complete this step.



I created the RMA table in the QuantigrationUpdates database based on the ERD. The table includes RMAID as the primary key and OrderID as a foreign key to establish a relationship with the Orders table. This ensures that each RMA record is associated with a valid order. Additional attributes such as step, status, and reason were included to store RMA process information. I verified the table creation using the SHOW TABLES and DESCRIBE RMA; commands to confirm that the table and its structure were created correctly.

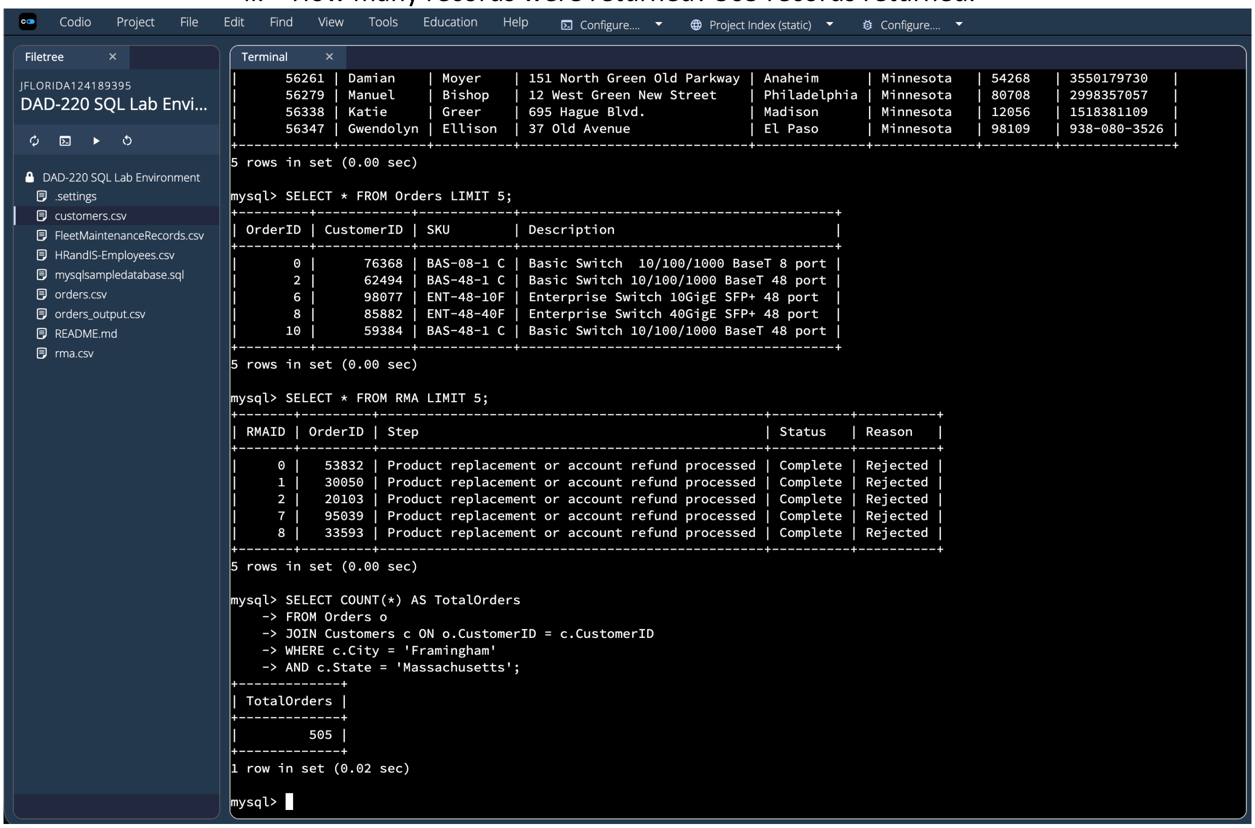
Step Two: Load and Query the Data

1. **Import** the **data** from each file **into tables.**
    a. Use the QuantigrationUpdates database, the three tables you created, and the three CSV files preloaded into Codio.
    b. Use the import utility of your database program to load the data from each file into the table of the same name. Perform this step three times, once for each table.
    c. Provide the SQL commands you ran against MySQL to complete this step.



I imported the data from the provided CSV files into the Customers, Orders, and RMA tables using the LOAD DATA INFILE method in MySQL. Each file was loaded into its corresponding table with fields separated by commas and lines terminated by \r\n. Due to the large volume of data, I verified the import by running summary queries such as SELECT COUNT(*) and sample queries using LIMIT to confirm that records were successfully inserted into each table.

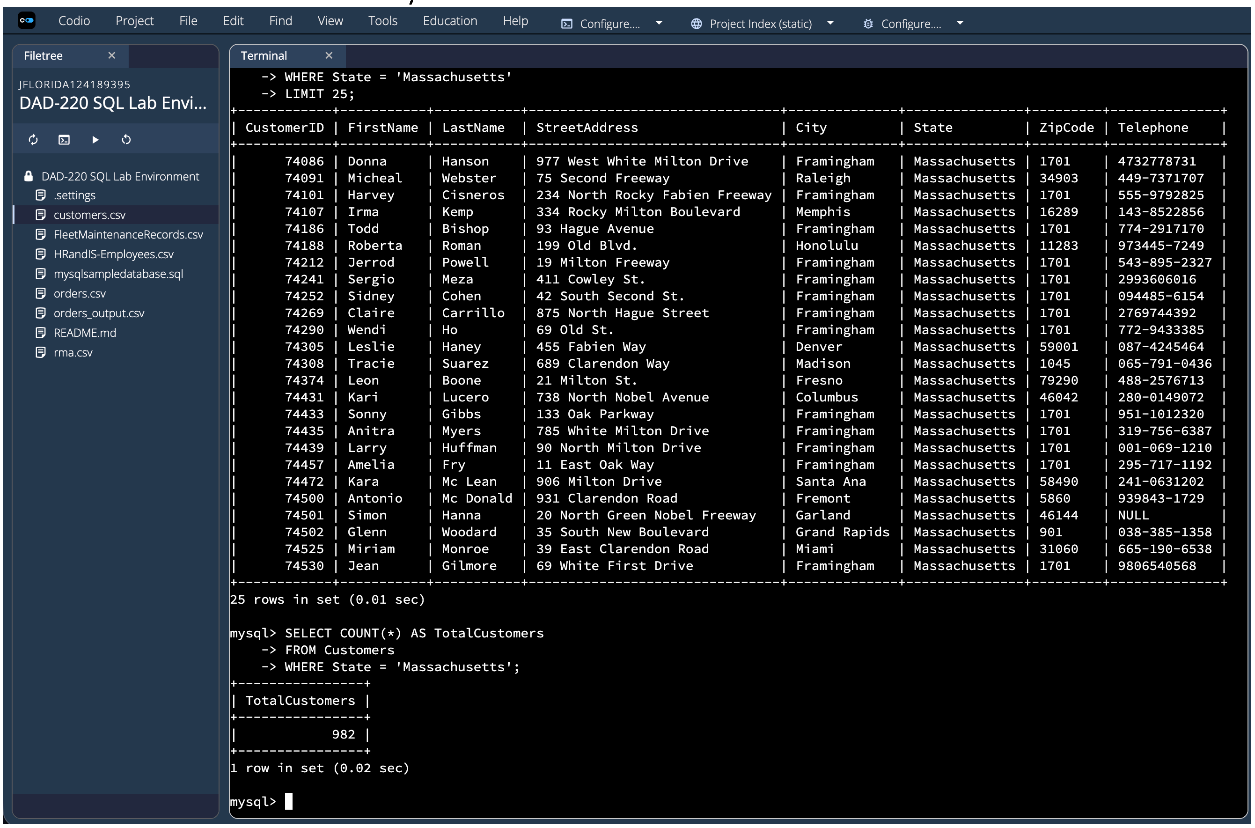
2. **Write basic queries** against the imported tables to organize and analyze the targeted data**.** For each query, replace the bracketed text with a screenshot of the query and its output. Also, include a one- to three-sentence description of the output.
    a. Write a SQL query that returns the count of orders for customers located only in Framingham, Massachusetts.
        i. This query will use a table join between the Customers and Orders tables. The query will also use a WHERE clause.
        ii. How many records were returned? 505 records returned.




```
|      56261 | Damian     | Moyer    | 151 North Green Old Parkway | Anaheim      | Minnesota    | 54268    | 3550179730   |
|      56279 | Manuel     | Bishop   | 12 West Green New Street    | Philadelphia | Minnesota    | 80708    | 2998357057   |
|      56338 | Katie      | Greer    | 695 Hague Blvd.             | Madison      | Minnesota    | 12056    | 1518381109   |
|      56347 | Gwendolyn  | Ellison  | 37 Old Avenue               | El Paso      | Minnesota    | 98109    | 938-080-3526 |
+------------+------------+----------+-----------------------------+--------------+--------------+----------+--------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Orders LIMIT 5;
+---------+------------+------------+------------------------------------------+
| OrderID | CustomerID | SKU        | Description                              |
+---------+------------+------------+------------------------------------------+
|       0 |      76368 | BAS-08-1 C | Basic Switch  10/100/1000 BaseT 8 port   |
|       2 |      62494 | BAS-48-1 C | Basic Switch 10/100/1000 BaseT 48 port   |
|       6 |      98077 | ENT-48-10F | Enterprise Switch 10GigE SFP+ 48 port    |
|       8 |      85882 | ENT-48-40F | Enterprise Switch 40GigE SFP+ 48 port    |
|      10 |      59384 | BAS-48-1 C | Basic Switch 10/100/1000 BaseT 48 port   |
+---------+------------+------------+------------------------------------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM RMA LIMIT 5;
+-------+---------+--------------------------------------------------+----------+----------+
| RMAID | OrderID | Step                                             | Status   | Reason   |
+-------+---------+--------------------------------------------------+----------+----------+
|     0 |   53832 | Product replacement or account refund processed | Complete | Rejected |
|     1 |   30050 | Product replacement or account refund processed | Complete | Rejected |
|     2 |   20103 | Product replacement or account refund processed | Complete | Rejected |
|     7 |   95039 | Product replacement or account refund processed | Complete | Rejected |
|     8 |   33593 | Product replacement or account refund processed | Complete | Rejected |
+-------+---------+--------------------------------------------------+----------+----------+
5 rows in set (0.00 sec)

mysql> SELECT COUNT(*) AS TotalOrders
    -> FROM Orders o
    -> JOIN Customers c ON o.CustomerID = c.CustomerID
    -> WHERE c.City = 'Framingham'
    -> AND c.State = 'Massachusetts';
+-------------+
| TotalOrders |
+-------------+
|         505 |
+-------------+
1 row in set (0.02 sec)

mysql> 
```

I wrote a SQL query to count the number of orders made by customers located in Framingham, Massachusetts. This query uses a JOIN between the Orders and Customers tables based on the CustomerID field. A WHERE clause was used to filter the records by city and state, and the COUNT function was used to return the total number of matching orders of 505 records.

b. Write a SQL query to select all of the customers located in Massachusetts.
    i. Use a WHERE clause to limit the number of records in the Customers table to only those who are located in Massachusetts.
    ii. How many records were returned? 982 records were returned.



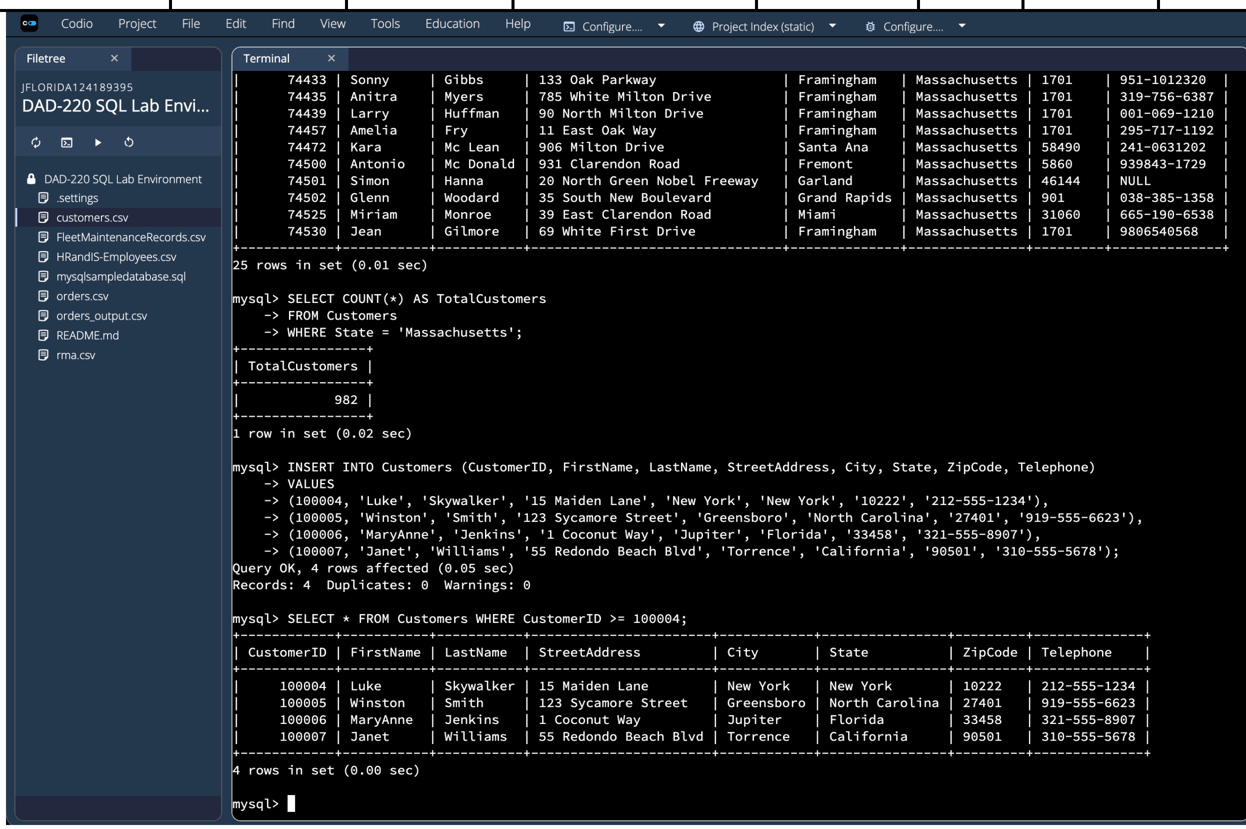

```
    -> WHERE State = 'Massachusetts'
    -> LIMIT 25;
+------------+-----------+-----------+----------------------------+-------------+---------------+---------+--------------+
| CustomerID | FirstName | LastName  | StreetAddress              | City        | State         | ZipCode | Telephone    |
+------------+-----------+-----------+----------------------------+-------------+---------------+---------+--------------+
|      74086 | Donna     | Hanson    | 977 West White Milton Drive | Framingham | Massachusetts | 1701    | 4732778731   |
|      74091 | Micheal   | Webster   | 75 Second Freeway          | Raleigh     | Massachusetts | 34903   | 449-7371707  |
|      74101 | Harvey    | Cisneros  | 234 North Rocky Fabien Freeway | Framingham | Massachusetts | 1701 | 555-9792825  |
|      74107 | Irma      | Kemp      | 334 Rocky Milton Boulevard | Memphis     | Massachusetts | 16289   | 143-8522856  |
|      74186 | Todd      | Bishop    | 93 Hague Avenue            | Framingham  | Massachusetts | 1701    | 774-2917170  |
|      74188 | Roberta   | Roman     | 199 Old Blvd.              | Honolulu    | Massachusetts | 11283   | 973445-7249  |
|      74212 | Jerrod    | Powell    | 19 Milton Freeway          | Framingham  | Massachusetts | 1701    | 543-895-2327 |
|      74241 | Sergio    | Meza      | 411 Cowley St.             | Framingham  | Massachusetts | 1701    | 2993606016   |
|      74252 | Sidney    | Cohen     | 42 South Second St.        | Framingham  | Massachusetts | 1701    | 094485-6154  |
|      74269 | Claire    | Carrillo  | 875 North Hague Street     | Framingham  | Massachusetts | 1701    | 2769744392   |
|      74290 | Wendi     | Ho        | 69 Old St.                 | Framingham  | Massachusetts | 1701    | 772-9433385  |
|      74305 | Leslie    | Haney     | 455 Fabien Way             | Denver      | Massachusetts | 59001   | 087-4245464  |
|      74308 | Tracie    | Suarez    | 689 Clarendon Way          | Madison     | Massachusetts | 1045    | 065-791-0436 |
|      74374 | Leon      | Boone     | 21 Milton St.              | Fresno      | Massachusetts | 79290   | 488-2576713  |
|      74431 | Kari      | Lucero    | 738 North Nobel Avenue     | Columbus    | Massachusetts | 46042   | 280-0149072  |
|      74433 | Sonny     | Gibbs     | 133 Oak Parkway            | Framingham  | Massachusetts | 1701    | 951-1012320  |
|      74435 | Anitra    | Myers     | 785 White Milton Drive     | Framingham  | Massachusetts | 1701    | 319-756-6387 |
|      74439 | Larry     | Huffman   | 90 North Milton Drive      | Framingham  | Massachusetts | 1701    | 001-069-1210 |
|      74457 | Amelia    | Fry       | 11 East Oak Way            | Framingham  | Massachusetts | 1701    | 295-717-1192 |
|      74472 | Kara      | Mc Lean   | 906 Milton Drive           | Santa Ana   | Massachusetts | 58490   | 241-0631202  |
|      74500 | Antonio   | Mc Donald | 931 Clarendon Road         | Fremont     | Massachusetts | 5860    | 939843-1729  |
|      74501 | Simon     | Hanna     | 20 North Green Nobel Freeway | Garland   | Massachusetts | 46144   | NULL         |
|      74502 | Glenn     | Woodard   | 35 South New Boulevard     | Grand Rapids | Massachusetts | 901    | 038-385-1358 |
|      74525 | Miriam    | Monroe    | 39 East Clarendon Road     | Miami       | Massachusetts | 31060   | 665-190-6538 |
|      74530 | Jean      | Gilmore   | 69 White First Drive       | Framingham  | Massachusetts | 1701    | 9806540568   |
+------------+-----------+-----------+----------------------------+-------------+---------------+---------+--------------+
25 rows in set (0.01 sec)

mysql> SELECT COUNT(*) AS TotalCustomers
    -> FROM Customers
    -> WHERE State = 'Massachusetts';
+----------------+
| TotalCustomers |
+----------------+
|            982 |
+----------------+
1 row in set (0.02 sec)

mysql>
```

I wrote a SQL query to count the number of customers located in Massachusetts. The query uses a WHERE clause to filter records in the Customers table based on the state value. The COUNT function was used to return the total number of matching records.

c. Write a SQL query to insert four new records into the Orders and Customers tables using the data below:

Customers Table

CustomerID	FirstName	LastName	StreetAddress	City	State	ZipCode	Telephone
100004	Luke	Skywalker	15 Maiden Lane	New York	NY	10222	212-555-1234
100005	Winston	Smith	123 Sycamore Street	Greensboro	NC	27401	919-555-6623
100006	MaryAnne	Jenkins	1 Coconut Way	Jupiter	FL	33458	321-555-8907
100007	Janet	Williams	55 Redondo Beach Blvd	Torrence	CA	90501	310-555-5678

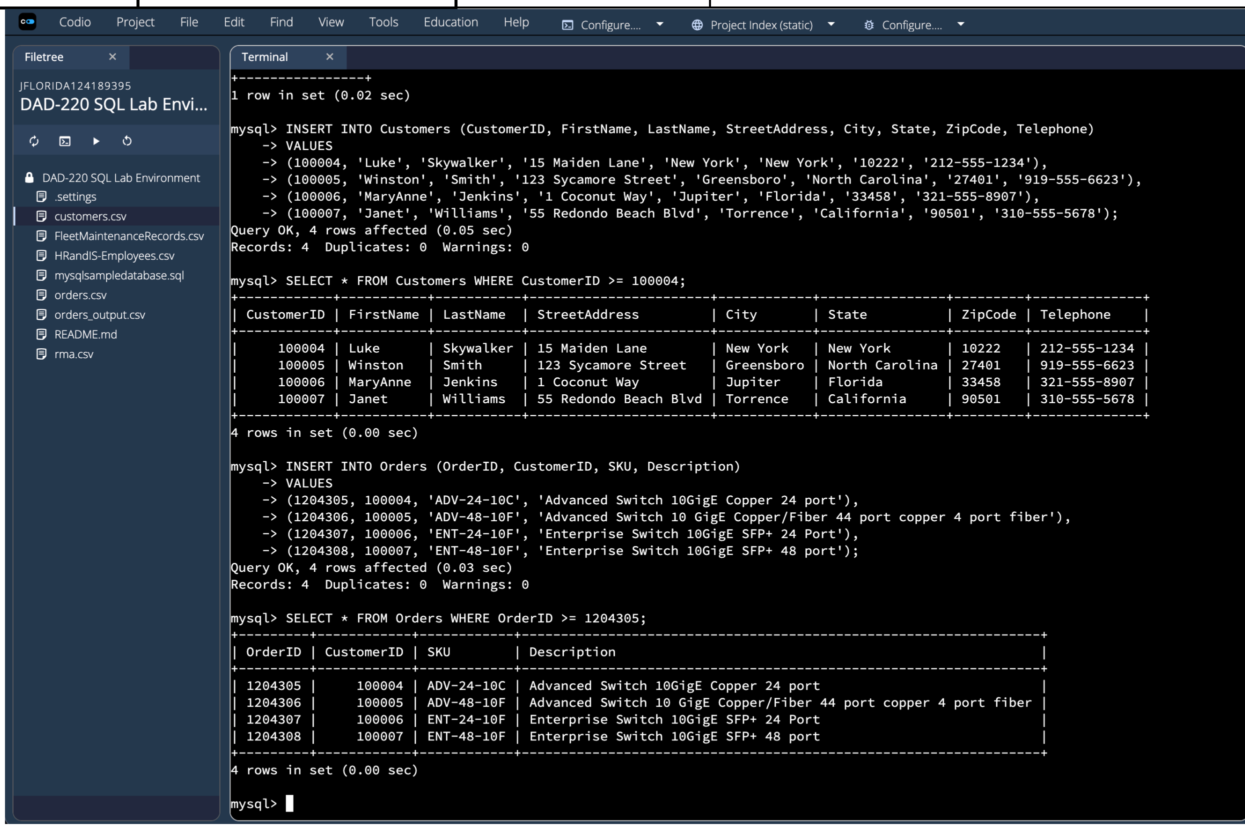


```
|      74433 | Sonny      | Gibbs      | 133 Oak Parkway                  | Framingham   | Massachusetts | 1701    | 951-1012320   |
|      74435 | Anitra     | Myers      | 785 White Milton Drive           | Framingham   | Massachusetts | 1701    | 319-756-6387  |
|      74439 | Larry      | Huffman    | 90 North Milton Drive            | Framingham   | Massachusetts | 1701    | 001-069-1210  |
|      74457 | Amelia     | Fry        | 11 East Oak Way                  | Framingham   | Massachusetts | 1701    | 295-717-1192  |
|      74472 | Kara       | Mc Lean    | 906 Milton Drive                 | Santa Ana    | Massachusetts | 58490   | 241-0631202   |
|      74500 | Antonio    | Mc Donald  | 931 Clarendon Road               | Fremont      | Massachusetts | 5860    | 939843-1729   |
|      74501 | Simon      | Hanna      | 20 North Green Nobel Freeway     | Garland      | Massachusetts | 46144   | NULL          |
|      74502 | Glenn      | Woodard    | 35 South New Boulevard           | Grand Rapids | Massachusetts | 901     | 038-385-1358  |
|      74525 | Miriam     | Monroe     | 39 East Clarendon Road           | Miami        | Massachusetts | 31060   | 665-190-6538  |
|      74530 | Jean       | Gilmore    | 69 White First Drive             | Framingham   | Massachusetts | 1701    | 9806540568    |
+------------+------------+------------+----------------------------------+--------------+---------------+---------+---------------+
25 rows in set (0.01 sec)

mysql> SELECT COUNT(*) AS TotalCustomers
    -> FROM Customers
    -> WHERE State = 'Massachusetts';
+----------------+
| TotalCustomers |
+----------------+
|            982 |
+----------------+
1 row in set (0.02 sec)

mysql> INSERT INTO Customers (CustomerID, FirstName, LastName, StreetAddress, City, State, ZipCode, Telephone)
    -> VALUES
    -> (100004, 'Luke', 'Skywalker', '15 Maiden Lane', 'New York', 'New York', '10222', '212-555-1234'),
    -> (100005, 'Winston', 'Smith', '123 Sycamore Street', 'Greensboro', 'North Carolina', '27401', '919-555-6623'),
    -> (100006, 'MaryAnne', 'Jenkins', '1 Coconut Way', 'Jupiter', 'Florida', '33458', '321-555-8907'),
    -> (100007, 'Janet', 'Williams', '55 Redondo Beach Blvd', 'Torrence', 'California', '90501', '310-555-5678');
Query OK, 4 rows affected (0.05 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Customers WHERE CustomerID >= 100004;
+------------+-----------+-----------+-----------------------+------------+----------------+---------+--------------+
| CustomerID | FirstName | LastName  | StreetAddress         | City       | State          | ZipCode | Telephone    |
+------------+-----------+-----------+-----------------------+------------+----------------+---------+--------------+
|     100004 | Luke      | Skywalker | 15 Maiden Lane        | New York   | New York       | 10222   | 212-555-1234 |
|     100005 | Winston   | Smith     | 123 Sycamore Street   | Greensboro | North Carolina | 27401   | 919-555-6623 |
|     100006 | MaryAnne  | Jenkins   | 1 Coconut Way         | Jupiter    | Florida        | 33458   | 321-555-8907 |
|     100007 | Janet     | Williams  | 55 Redondo Beach Blvd | Torrence   | California     | 90501   | 310-555-5678 |
+------------+-----------+-----------+-----------------------+------------+----------------+---------+--------------+
4 rows in set (0.00 sec)

mysql>
```

I inserted four new records into the Customers table using the INSERT INTO statement based on the data provided in the assignment. Each record includes customer details such as name, address, city, state, zip code, and telephone number. I verified that the records were successfully added by running a SELECT query to display the newly inserted customer entries.

Orders Table

OrderID	CustomerID	SKU	Description
1204305	100004	ADV-24-10C	Advanced Switch 10GigE Copper 24 port
1204306	100005	ADV-48-10F	Advanced Switch 10 GigE Copper/Fiber 44 port copper 4 port fiber
1204307	100006	ENT-24-10F	Enterprise Switch 10GigE SFP+ 24 Port
1204308	100007	ENT-48-10F	Enterprise Switch 10GigE SFP+ 48 port

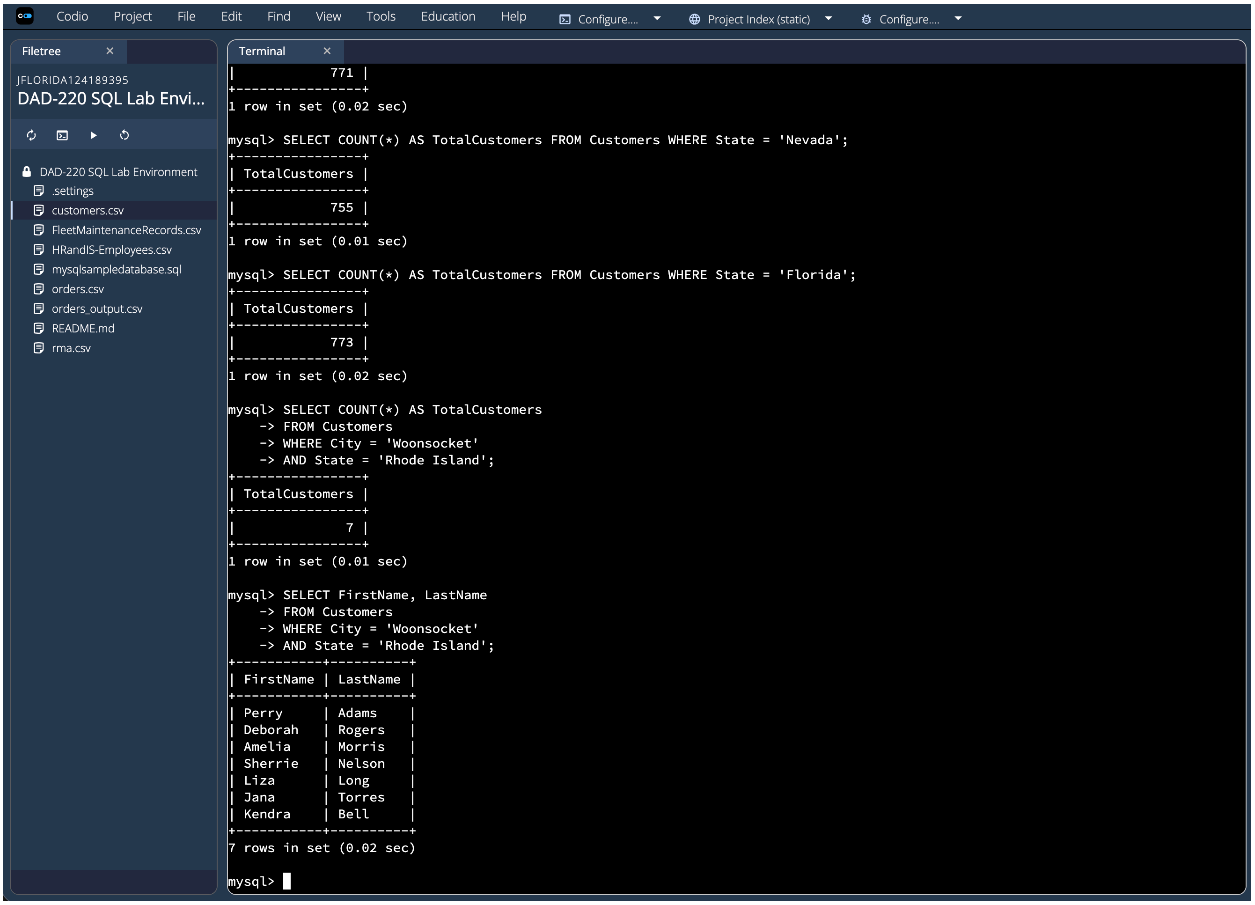



```
+-----------------+
1 row in set (0.02 sec)

mysql> INSERT INTO Customers (CustomerID, FirstName, LastName, StreetAddress, City, State, ZipCode, Telephone)
    -> VALUES
    -> (100004, 'Luke', 'Skywalker', '15 Maiden Lane', 'New York', 'New York', '10222', '212-555-1234'),
    -> (100005, 'Winston', 'Smith', '123 Sycamore Street', 'Greensboro', 'North Carolina', '27401', '919-555-6623'),
    -> (100006, 'MaryAnne', 'Jenkins', '1 Coconut Way', 'Jupiter', 'Florida', '33458', '321-555-8907'),
    -> (100007, 'Janet', 'Williams', '55 Redondo Beach Blvd', 'Torrence', 'California', '90501', '310-555-5678');
Query OK, 4 rows affected (0.05 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Customers WHERE CustomerID >= 100004;
+------------+-----------+-----------+-----------------------+------------+----------------+---------+--------------+
| CustomerID | FirstName | LastName  | StreetAddress         | City       | State          | ZipCode | Telephone    |
+------------+-----------+-----------+-----------------------+------------+----------------+---------+--------------+
|     100004 | Luke      | Skywalker | 15 Maiden Lane        | New York   | New York       | 10222   | 212-555-1234 |
|     100005 | Winston   | Smith     | 123 Sycamore Street   | Greensboro | North Carolina | 27401   | 919-555-6623 |
|     100006 | MaryAnne  | Jenkins   | 1 Coconut Way         | Jupiter    | Florida        | 33458   | 321-555-8907 |
|     100007 | Janet     | Williams  | 55 Redondo Beach Blvd | Torrence   | California     | 90501   | 310-555-5678 |
+------------+-----------+-----------+-----------------------+------------+----------------+---------+--------------+
4 rows in set (0.00 sec)

mysql> INSERT INTO Orders (OrderID, CustomerID, SKU, Description)
    -> VALUES
    -> (1204305, 100004, 'ADV-24-10C', 'Advanced Switch 10GigE Copper 24 port'),
    -> (1204306, 100005, 'ADV-48-10F', 'Advanced Switch 10 GigE Copper/Fiber 44 port copper 4 port fiber'),
    -> (1204307, 100006, 'ENT-24-10F', 'Enterprise Switch 10GigE SFP+ 24 Port'),
    -> (1204308, 100007, 'ENT-48-10F', 'Enterprise Switch 10GigE SFP+ 48 port');
Query OK, 4 rows affected (0.03 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Orders WHERE OrderID >= 1204305;
+---------+------------+------------+------------------------------------------------------------------+
| OrderID | CustomerID | SKU        | Description                                                      |
+---------+------------+------------+------------------------------------------------------------------+
| 1204305 |     100004 | ADV-24-10C | Advanced Switch 10GigE Copper 24 port                            |
| 1204306 |     100005 | ADV-48-10F | Advanced Switch 10 GigE Copper/Fiber 44 port copper 4 port fiber |
| 1204307 |     100006 | ENT-24-10F | Enterprise Switch 10GigE SFP+ 24 Port                            |
| 1204308 |     100007 | ENT-48-10F | Enterprise Switch 10GigE SFP+ 48 port                            |
+---------+------------+------------+------------------------------------------------------------------+
4 rows in set (0.00 sec)

mysql>
```

I inserted four new records into the Orders table using the INSERT INTO statement based on the data provided in the assignment. Each record includes an OrderID, CustomerID, SKU, and description. The CustomerID values were used to maintain the relationship between the Orders and Customers tables through the foreign key. I verified that the records were successfully added by running a SELECT query to display the newly inserted order entries.

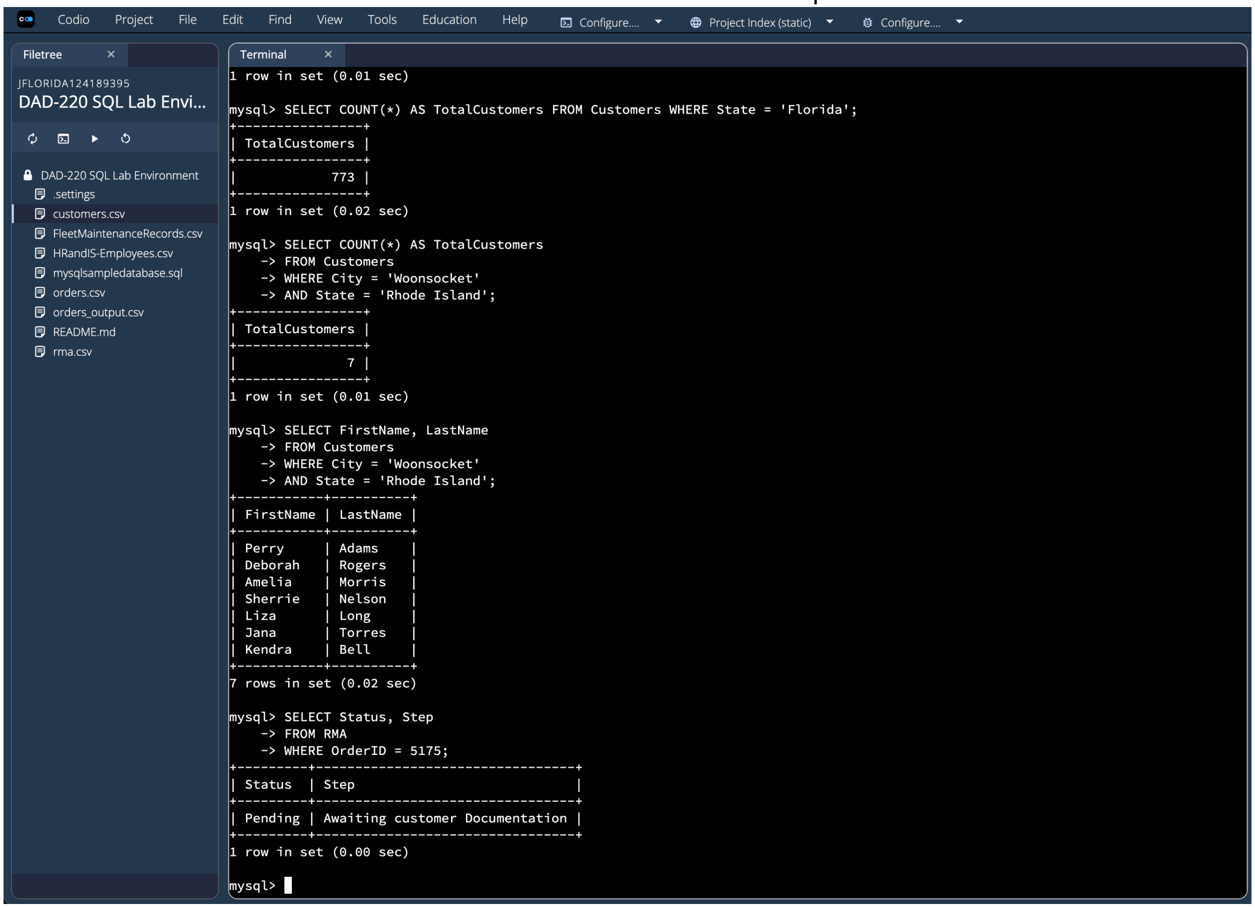
d. In the Customers table, perform a query to count all records where the city is Woonsocket and the state is Rhode Island.

    i. How many records are in the Customers table where the field "city" equals "Woonsocket"? 7 records were returned.



I wrote a SQL query to count the number of customers located in Woonsocket, Rhode Island. The query uses a WHERE clause to filter records in the Customers table based on both the city and state values. The COUNT function was used to return the total number of matching records, and I also verified the result by displaying the names of the customers that meet the criteria.

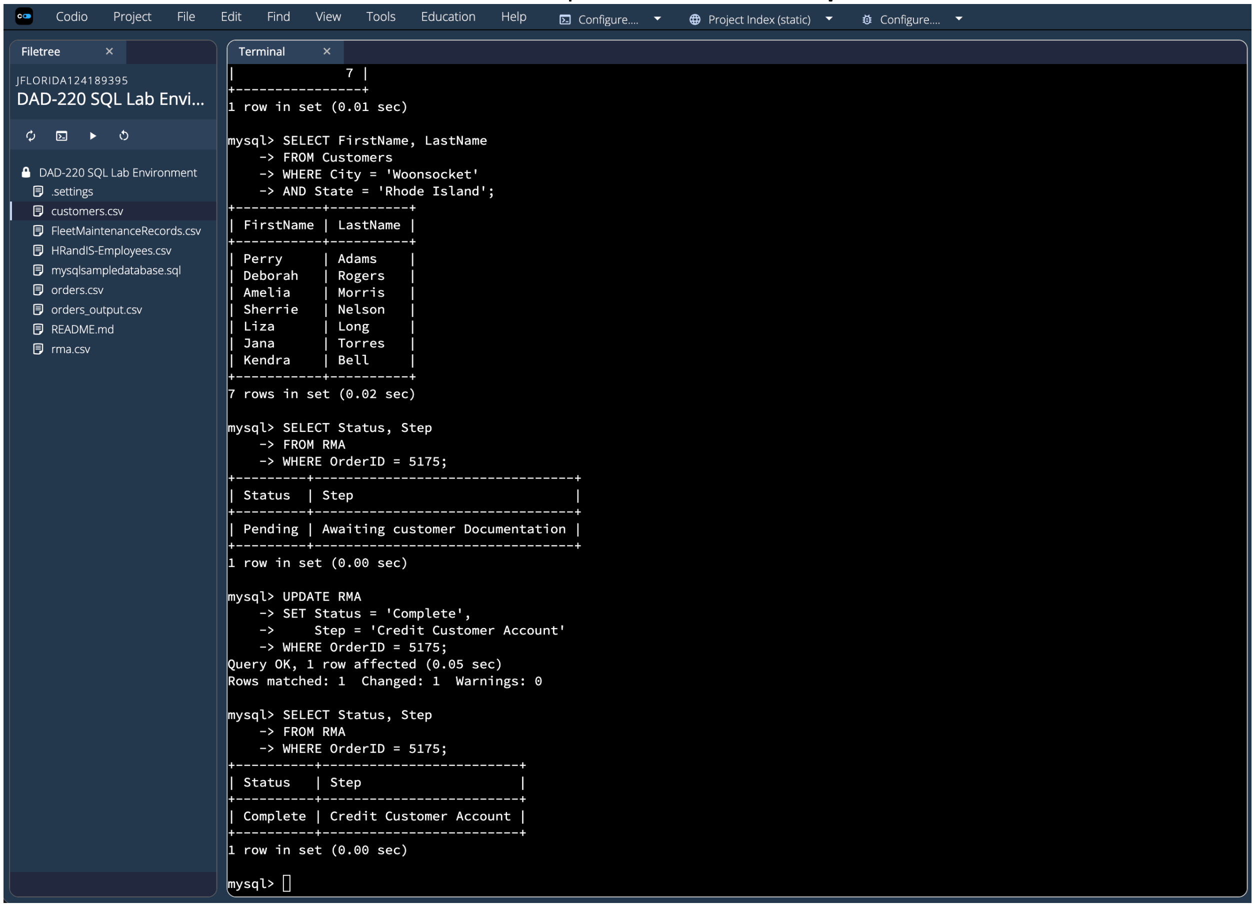
e. In the RMA database, update a customer's records.
   i. Write a SQL statement to select the current fields of status and step for the record in the RMA table with an OrderID value of "5175".
      1. What are the current status and step?



I wrote a SQL query to retrieve the current status and step for the RMA record with an OrderID of 5175. The query uses a WHERE clause to filter the specific record in the RMA table. This allowed me to identify the current values of the status and step fields before performing any updates.

ii. Write a SQL statement to update the **status** and **step** for the **OrderID**, 5175 to **status** = "Complete" and **step** = "Credit Customer Account".

1. What are the updated **status** and **step** values for this record?

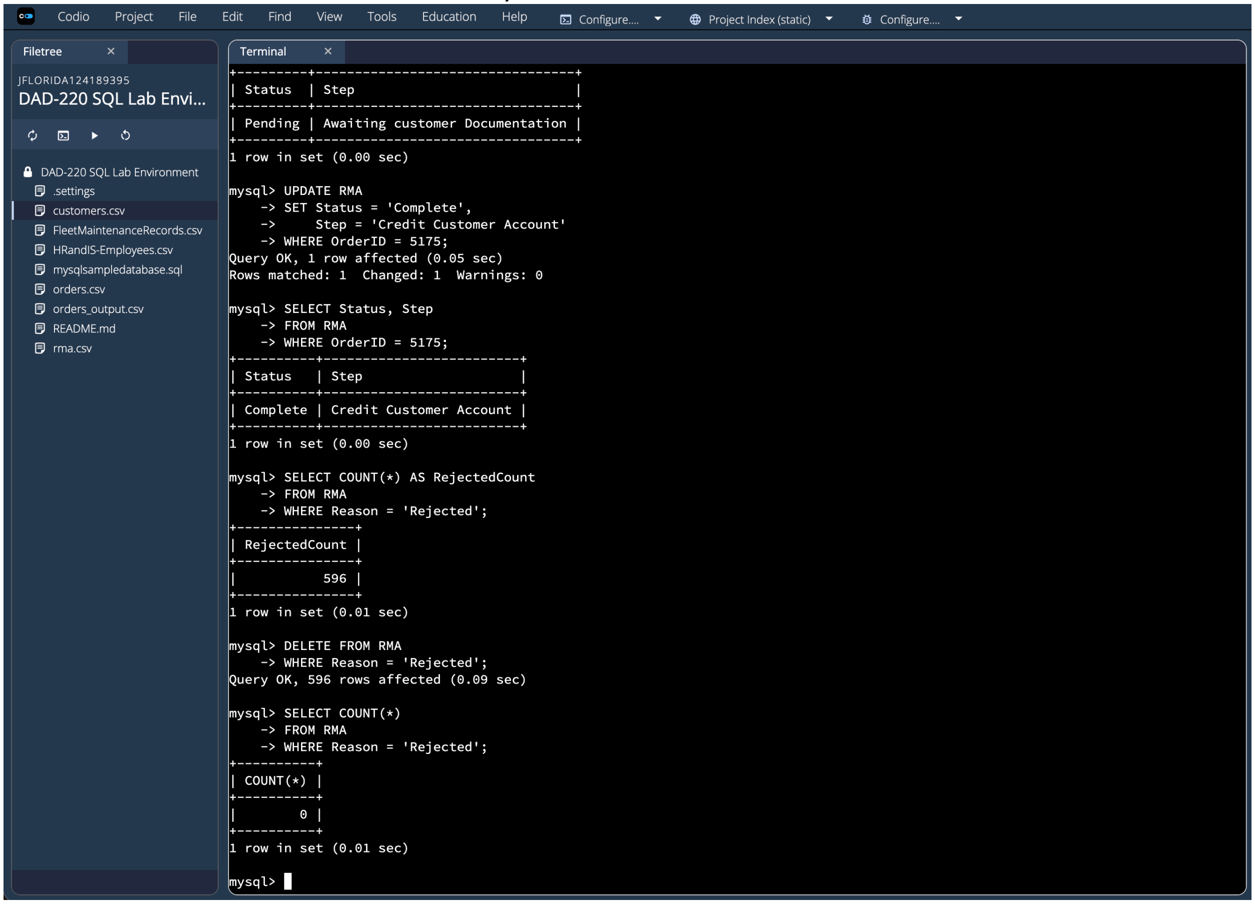


I used an UPDATE statement to modify the existing record in the RMA table for the specified OrderID 5175. The status field was changed from "Pending" to "Complete," and the step field was updated to "Credit Customer Account." After performing the update, I ran a SELECT query to verify that the changes were successfully applied and confirmed that the record now reflects the updated values.

f. Delete RMA records.

i. Write a SQL statement to delete all records with a reason of "Rejected".

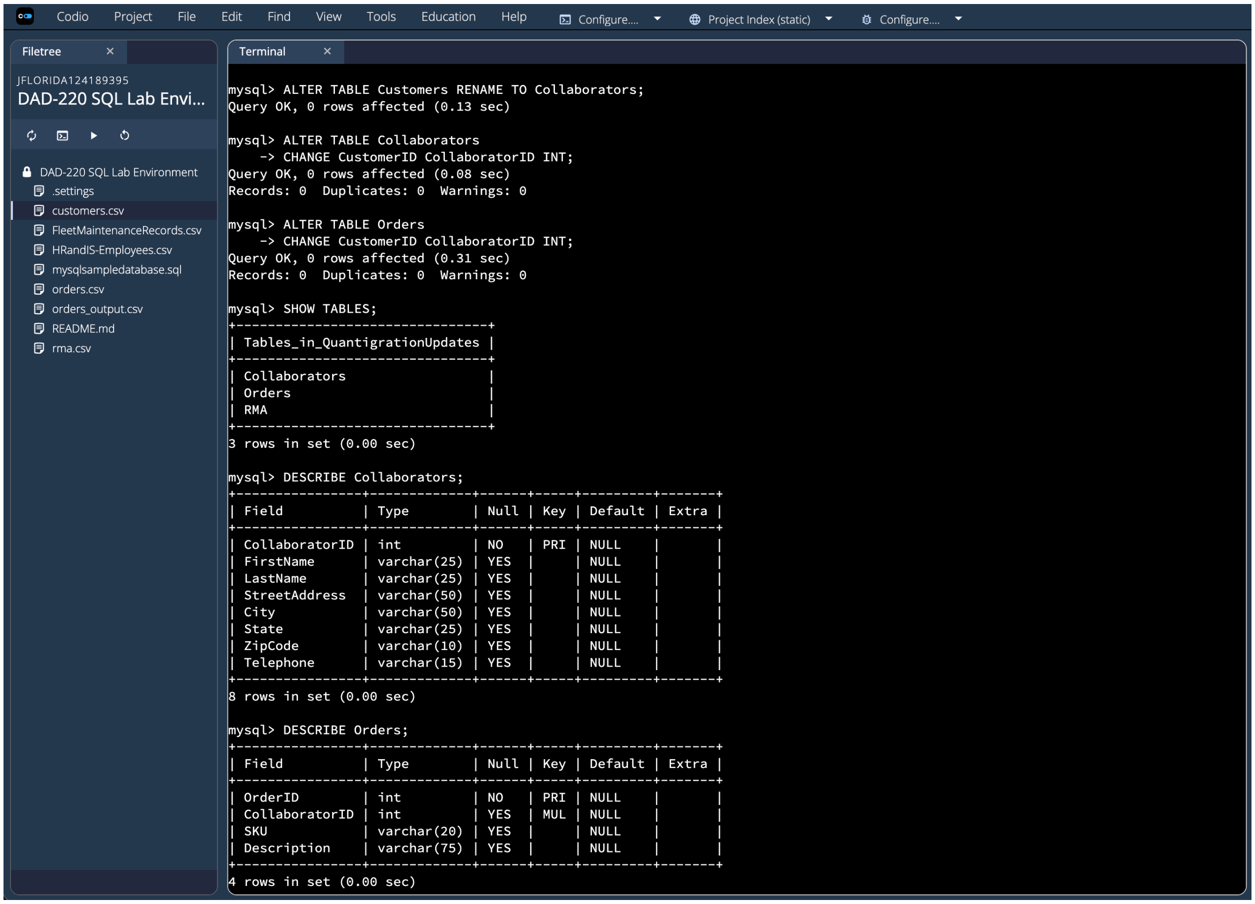
1. How many records were deleted? 596 records were deleted.



I used a SELECT COUNT(*) query to identify how many records in the RMA table had a reason of "Rejected," which returned 596 records. After determining the number of affected records, I executed a DELETE statement to remove all records with that condition. I then verified the deletion by running another SELECT query, which confirmed that the number of records with the reason "Rejected" is now 0, indicating that all matching records were successfully removed.

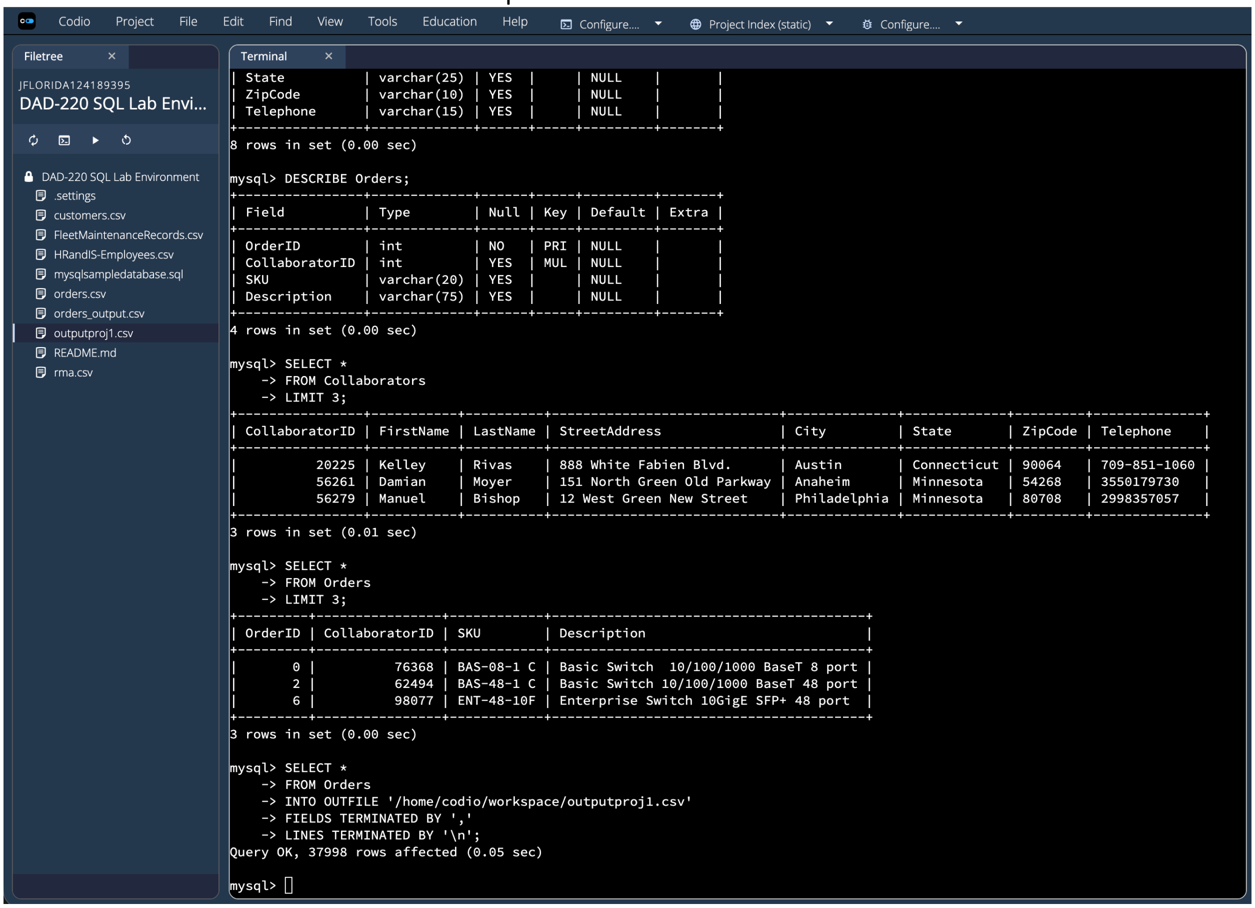
snhu

3. **Update your existing tables** from "Customer" to "Collaborator" using SQL based on this change in requirements. Copy and paste the SQL you write to do the following action:
    a. Rename all instances of "Customer" to "Collaborator".



I used ALTER TABLE statements to update the database schema based on the new naming requirement. The Customers table was successfully renamed to Collaborators, and the CustomerID column was changed to CollaboratorID. I also updated the Orders table by renaming its foreign key column to maintain consistency between related tables. I verified the changes using SHOW TABLES and DESCRIBE statements, confirming that the table and column names were updated correctly while preserving the data and relationships.

4. **Create** an **output file** of the required query results. Write a SQL statement to list the contents of the **Orders** table and send the output to a file that has a CSV extension.



I used a SELECT statement with the INTO OUTFILE clause to export the contents of the Orders table into a CSV file named outputproj1.csv. The query specified comma-separated values using FIELDS TERMINATED BY and defined each row using LINES TERMINATED BY. The output confirmed that 37,998 rows were successfully written to the file, and the file was created in the workspace, verifying that the data export was completed correctly.